MAIS PAC

IPhO 02.14: Structure of Matter and Nuclear Physics · Lecture + Problems

Structure of Matter and Nuclear Physics

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. The atom is mostly empty space

The discovery of the electron in 1897 left a puzzle: if negative electrons are inside a neutral atom, where is the positive charge? Thomson proposed a diffuse "plum pudding"; Rutherford's students fired alpha particles at gold foil and found that about one in eight thousand came straight back. A diffuse charge could never do that.

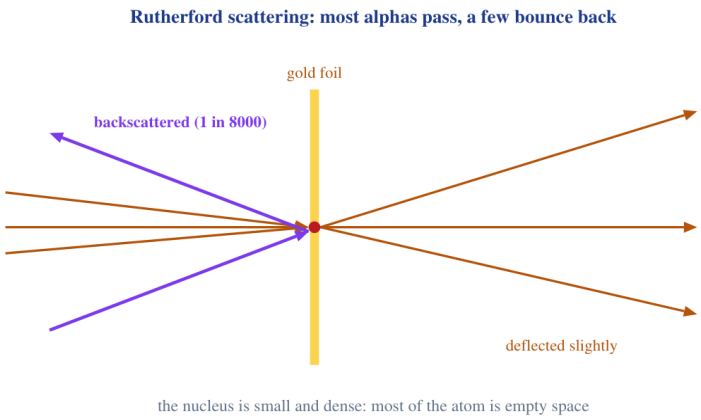


Figure 1. A diffuse positive charge could not turn an alpha particle round. Only a tiny, heavy, positively charged nucleus can, which is how the nuclear model was established in 1911.

Rutherford's conclusion is the nuclear model: a tiny, dense, positively charged **nucleus** of radius about 10^(-15) {m} holds almost all the mass, and the electrons occupy a volume about 10^5 times larger. The atom is 10^(-10) {m} across and is essentially empty.

particle	symbol	charge	mass
proton	^1_1{p}	+e	1.673×10^(-27) {kg}
neutron	^1_0{n}	0	1.675×10^(-27) {kg}
electron	^0_(-1){e}	-e	9.11×10^(-31) {kg}

A nuclide is written ^A_Z{X}: Z protons define the element, A-Z neutrons set the isotope, and A is the nucleon number. **Isotopes** have the same Z but different A — chemically identical, but with different nuclear masses and stability.

2. Inside the nucleon

Protons and neutrons are not elementary. They contain three **quarks**: up (+2/3,e) and down (-1/3,e).

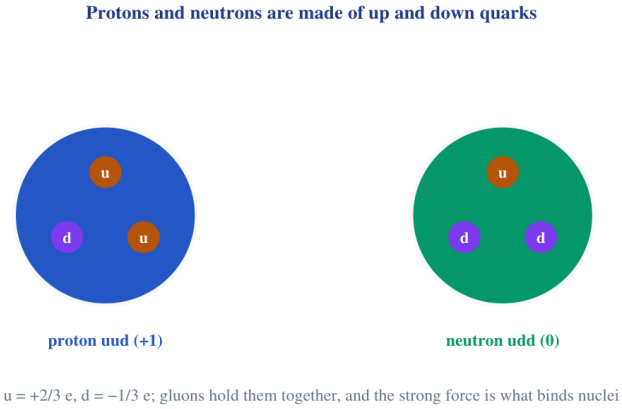


Figure 4. The quark charges add to the observed nucleon charges: $2/3+2/3-1/3 = +1$ for the proton, and $2/3-1/3-1/3 = 0$ for the neutron. Quarks never appear alone.

$$\{proton\} = uud: (2)/(3)+(2)/(3)-(1)/(3) = +1, \{neutron\} = udd: (2)/(3)-(1)/(3)-(1)/(3) = 0. \quad (1)$$

The quarks are held by **gluons** through the strong interaction, and the same force, leaking out of the nucleons, binds nuclei together. Quarks never appear in isolation: pulling them apart creates new quark–antiquark pairs instead, which is why the only free particles we see are colour-neutral combinations.

3. Mass defect and binding energy

The mass of a nucleus is **less** than the sum of the masses of its separate nucleons. The difference, the **mass defect** Δ m, is the energy that was released when the nucleus formed:

$$E_b = \Delta m,c^2 = [Zm_p + (A-Z)m_n - m_(\{nucleus\})]c^2. \quad (2)$$

It is convenient to measure masses in atomic mass units, 1 {u} = 931.5 {MeV}/c², so that the energy in MeV is just the mass defect in {u} times 931.5.

The quantity that decides nuclear stability is the **binding energy per nucleon**, E_b/A.

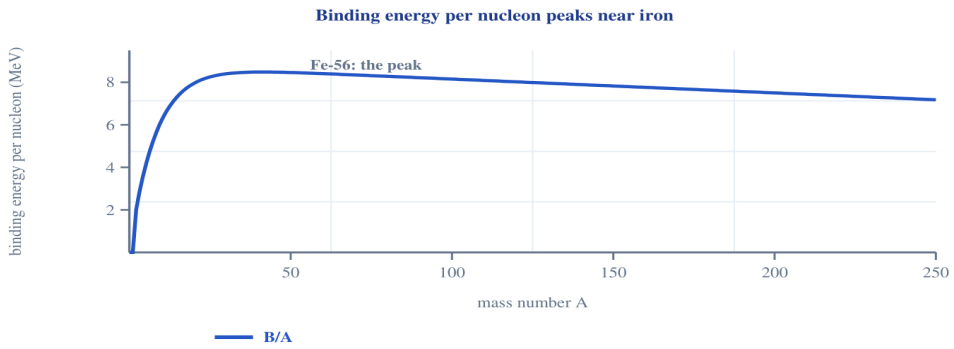


Figure. Fusion of light nuclei and fission of heavy ones both climb towards the peak, and both release energy.

- It rises steeply for the lightest nuclei, peaks at about 8.8 {MeV} near iron-56, and then falls slowly to about 7.6 {MeV} for uranium.
- **Fusion** of light nuclei moves products up the curve and releases energy — this powers stars.
- **Fission** of heavy nuclei moves fragments up the curve and also releases energy — this powers reactors.

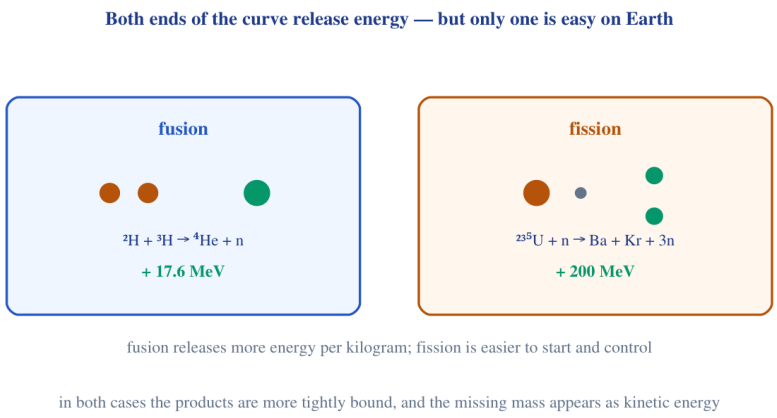


Figure 3. The energy comes from the mass difference: the products are lighter than the reactants by about 0.1% (fission) to 0.4% (fusion) of the total mass, and $E = \Delta mc^2$.

4. Radioactivity

An unstable nucleus decays spontaneously. Three modes matter:

- **Alpha decay:** ^A_Z{X} → {}^(A-4)_(Z-2){Y} + {}^4_2{He}. The alpha is heavy and charged, so alpha radiation is easily stopped (by paper or a few centimetres of air) but does great damage if inhaled.
- **Beta minus decay:** ^A_Z{X} → {}^A_(Z+1){Y} + e^- + _e. A neutron turns into a proton; the antineutrino carries away the missing energy, which is why the beta spectrum is continuous.
- **Gamma emission:** an excited nucleus drops to a lower state, emitting a photon of MeV energy. Z and A are unchanged.

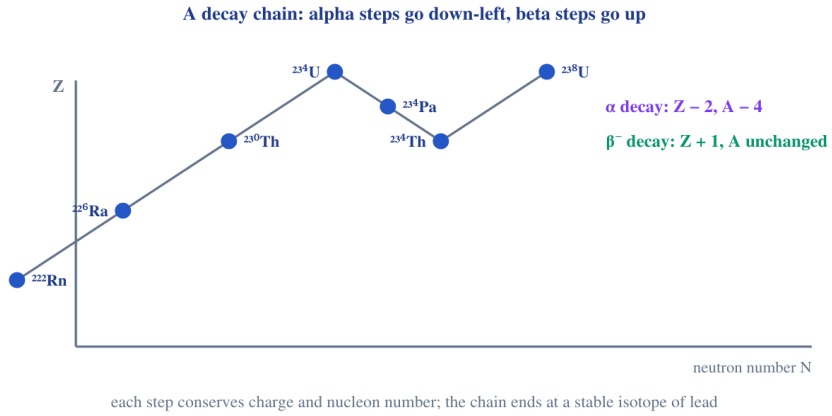


Figure 2. Alpha decay removes two protons and two neutrons, so it moves a nuclide down and to the left; beta decay turns a neutron into a proton, moving it up. The chain zigzags towards the line of stability.

Every decay conserves charge, nucleon number, energy and momentum, and the rules for the first two make nuclear equations easy to balance.

5. The exponential law

Decay is a **random** process with a constant probability per unit time, so the number of nuclei obeys

$$(dN)/(dt) = -\lambda N \; N = N_0 e^{(-\lambda t)}, \; A = \lambda N = A_0 e^{(-\lambda t)}. \tag{3}$$

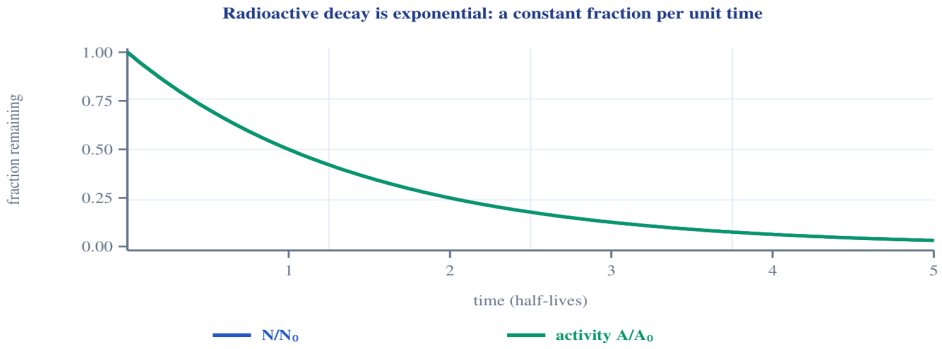


Figure. After each half-life half the nuclei remain, whatever the starting number.

Three related constants, often confused:

$$\{\text{decay constant}\} \; \lambda, \; \{\text{half-life}\} \; T_{1/2} = (\ln 2)/(\lambda), \; \{\text{mean lifetime}\} \; \tau = (1)/(\lambda). \tag{4}$$

Note $\tau = T_{1/2}/\ln 2 = 1.44, T_{1/2}$. The **activity** A is measured in becquerels (1 {Bq} = 1 decay per second); the older unit curie is 3.7×10^{10} {Bq}.

Because λ is constant, a sample never "runs out" at a definite time: the last few nuclei may survive for many half-lives, and the decay of a small sample looks visibly jagged.

Interactive experiment: Radioactive decay and activity — open the online handout to use it.

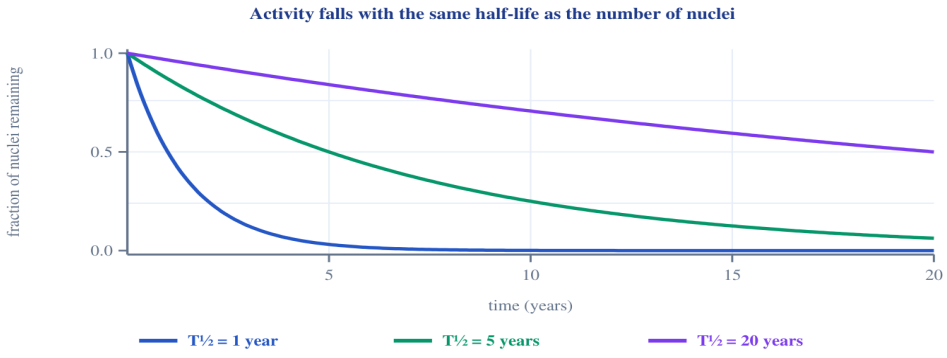


Figure. A short half-life means a high activity, and a sample decays away quickly.

6. Fission, fusion and radiation dose

Fission of ^{235}U releases about 200 {MeV} per event, fusion of deuterium and tritium about 17.6 {MeV} — but per kilogram, fusion releases several times more. Both are 10^6 times more energetic than chemical burning, because nuclear binding energies are of order MeV per nucleon while chemical energies are of order eV per atom.

Radiation dose is energy deposited per kilogram: the gray (1 {Gy} = 1 {J,kg^(-1)}) measures absorbed dose, and the sievert weights it by the biological damage of the radiation type. Natural background is about 2–3 {mSv} per year; a chest X-ray is about 0.1 {mSv}; 1 {Sv} in a short time causes acute illness.

The nuclear checklist. Balance Z and A before anything else — most "hard" nuclear equations become bookkeeping → convert mass to energy with 931.5 {MeV} per u (or c^2 = 9×10^(16) {J,kg^(-1)}) → for stability questions, ask whether E_b/A increases → for decay, use N = N_0e^(-λ t) with T_(1/2) = ln 2/λ and never confuse half-life with mean lifetime → remember that the activity depends on both λ and the number of nuclei present.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Nuclear notation. For the nuclide ^(235)_(92){U}: (a) state the number of protons, neutrons and nucleons; (b) find its mass in kilograms given its atomic mass is 235.04 {u}; (c) explain what is meant by an isotope and name one isotope of uranium.

Worked solution →

[1] Problem 2 — Balancing nuclear equations. Complete each equation and name the process: (a) ^(238)_(92){U} → {}^(234)_(90){Th} + ?; (b) ^(14)_6{C} → {}^(14)_7{N} + ?; (c) ²_1{H} + {}³_1{H} → {}^4_2{He} + ?.

Worked solution →

[2] Problem 3 — Mass defect of helium-4. The mass of a helium-4 nucleus is 4.0015 {u}, of a proton 1.00728 {u} and of a neutron 1.00866 {u}. (a) Find the mass defect. (b) Find the binding energy and the binding energy per nucleon. (c) Explain why the alpha particle is so tightly bound and what this means for alpha decay.

Worked solution →

[2] Problem 4 — Half-life and activity. A sample contains 4.0×10^(15) nuclei of a isotope with half-life 8.0 days. (a) Find the decay constant. (b) Find the initial activity in becquerels. (c) Find the activity after 24 days and the number of nuclei remaining.

Worked solution →

[2] Problem 5 — Carbon dating. Carbon-14 has a half-life of 5730 years and is present in living things at 1.3×10^(-12) of the carbon. A sample of wood has a specific activity of 1.5 {Bq,kg^(-1)} of carbon, compared with 2.3 {Bq,kg^(-1)} for living wood. (a) Find the age of the sample. (b) Explain the assumptions behind the method. (c) Explain why the method fails for samples older than about 50,000 years.

Worked solution →

[2] Problem 6 — Energy from fission. The fission of one ^(235){U} nucleus releases 200 {MeV}. (a) Find the energy released per fission in joules. (b) Find the energy released by 1.0 {kg} of ^(235){U}. (c) Find the mass converted to energy in (b) and express it as a fraction of the original mass.

Worked solution →

[2] Problem 7 — Alpha particle energy. An alpha particle is emitted from a nucleus with 5.5 {MeV} of kinetic energy. (a) Find its speed. (b) Find the speed of the recoiling nucleus if its mass is 220 {u}. (c) Explain why the alpha is stopped by a sheet of paper while gamma rays are not.

Worked solution →

[3] Problem 8 — The binding energy curve. Use the binding energies per nucleon 7.6 {MeV} for ^(238){U} and about 8.5 {MeV} for the fission fragments. (a) Estimate the energy released in one fission event. (b) Compare with 200 {MeV} and comment on the estimate. (c) Explain why fusion of hydrogen to helium releases energy per kilogram even more efficiently.

Worked solution →

[3] Problem 9 — A sealed source. A sealed source contains 2.0 {mg} of a nuclide of molar mass 60 {g,mol^(-1)} and half-life 5.3 years. (a) Find the number of atoms. (b) Find the activity. (c) Find the activity after 20 years and the total number of decays in the first 20 years.

Worked solution →

[3] Problem 10 — A nuclear reactor. A reactor produces 1.0 {GW} of thermal power from fission of ^(235){U}, releasing 200 {MeV} per fission, with a thermal efficiency of 33%. (a) Find the number of fissions per second and the fuel consumption in kg per day. (b) Find the electrical power output. (c) Explain why the fuel is enriched rather than natural uranium, and what a moderator does.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

PHYSNET / modern physics overview

Short modules on quanta, spectra, relativity and nuclear physics written for students.